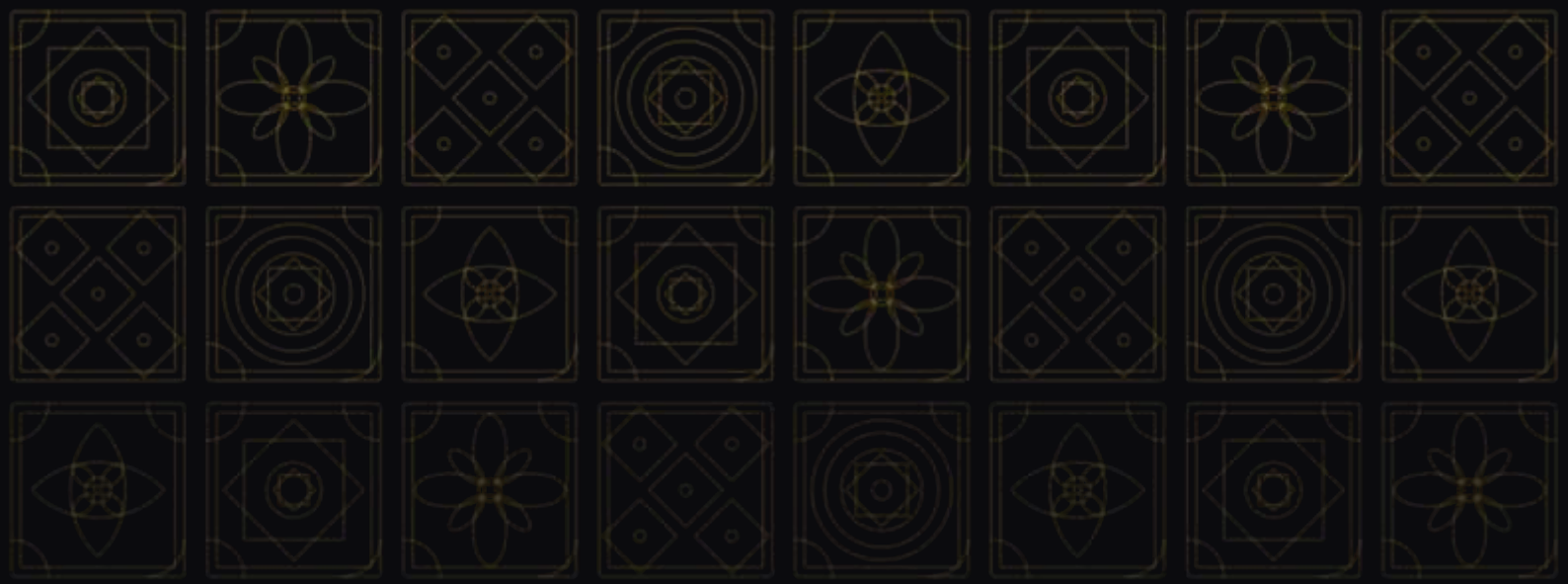




TRAINING PROGRAMME

AWS Cloud Practitioner

Understand, use and estimate the core services of Amazon Web Services

33 HOURS

FOR IT staff, developers, students and managers starting with AWS

PREPARES FOR AWS Certified Cloud Practitioner (CLF-C02)



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Overview

This programme gives participants a working knowledge of Amazon Web Services and prepares the AWS Certified Cloud Practitioner exam (CLF-C02). They learn the cloud models and AWS pricing, secure an account with IAM, then use the core services: networking with VPC, Route 53 and CloudFront, compute with EC2, Lambda and containers, storage, databases, scaling and monitoring, all framed by the Well-Architected Framework. Almost every chapter ends with a lab in the AWS console or CLI, exam practice closes the content, and the programme ends with a final project in which teams deploy a highly available web application and present its cost estimate.

At a glance

#	PHASE	FOCUS	HOURS
01	Cloud Concepts & Security	Cloud models, pricing and global infrastructure, IAM	6 h
02	Core Services	VPC, Route 53, CloudFront, EC2, Lambda, containers, storage, databases	9 h
03	Architecture, Operations & Exam Preparation	Well-Architected Framework, load balancing, Auto Scaling, CloudWatch, CLF-C02 practice	6 h
04	Final Project	A highly available web application and its cost estimate	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
	Total		33 h

Key facts

AUDIENCE	IT staff, developers, students and managers starting with AWS
PREREQUISITES	None. Basic IT literacy; networking or Linux basics help but are not required
DURATION	33 h: 27 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs in the AWS console and CLI; exam practice
LAB ENVIRONMENT	One laptop per participant with a browser and the AWS CLI; one AWS sandbox account per team, provided by the organiser, with a budget alert; resources destroyed at the end of each lab to control cost
ASSESSMENT	A lab validated at the end of each phase, practice exam questions and a final project presented to the group; certificate of completion for participants who complete the labs and the project
OUTCOME	Participants can explain and choose the core AWS services, deploy a simple highly available web application, estimate its cost and sit the CLF-C02 exam

PHASE 01

6 H

Cloud Concepts & Security

Participants learn what the cloud changes, how AWS charges for it and where it runs, then secure their sandbox account: the shared responsibility model, IAM users, groups, roles and policies, and MFA. Validated lab: an account with MFA on every user, an administrators group, a least-privilege role and a cost budget with an alert.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Cloud Concepts	Service and deployment models, benefits of the cloud	Lecture + Workshop	1.5 h
1.2	Economics & Global Infrastructure	Pricing models, total cost of ownership, billing, AWS Budgets and Cost Explorer, consolidated billing, support plans, Regions, Availability Zones, edge locations	Lecture + Lab	1.5 h
1.3	Security & IAM	Shared responsibility, users, groups, roles, policies, MFA, security services overview (GuardDuty, Inspector, Shield, WAF, AWS Artifact)	Lecture + Lab	3 h

PHASE 02

9 H

Core Services

Participants build with the services most workloads use: a VPC with public and private subnets, DNS and content delivery, virtual machines, functions and containers, object, block and file storage, and relational and NoSQL databases. Validated lab: a web server in a public subnet serving images from S3, with its database in a private subnet.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Networking & Content Delivery	VPC, subnets, security groups, Route 53, CloudFront	Lecture + Lab	3 h
2.2	Compute	EC2, Lambda, containers, Elastic Beanstalk	Lecture + Lab	3 h
2.3	Storage	S3 and its storage classes (including S3 Glacier), EBS, EFS	Lecture + Lab	1.5 h
2.4	Databases	RDS, Aurora, DynamoDB, Redshift	Lecture + Lab	1.5 h

PHASE 03

6 H

Architecture, Operations & Exam Preparation

Participants judge an architecture against the six pillars of the Well-Architected Framework, make it scale and heal with Elastic Load Balancing and Auto Scaling, watch it with CloudWatch, then practise exam questions across the four CLF-C02 domains. Validated lab: an Auto Scaling group behind a load balancer that replaces a stopped instance, with a CloudWatch alarm.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Cloud Architecture	Well-Architected Framework	Lecture + Workshop	1.5 h
3.2	Scaling & Monitoring	Elastic Load Balancing, Auto Scaling, CloudWatch	Lecture + Lab	1.5 h
3.3	CLF-C02 Preparation	Practice questions across the four domains; review of service categories not used in labs (analytics, AI/ML, migration)	Lecture + Workshop	3 h

PHASE 04

6 H

Final Project

Teams of two or three deploy a highly available web application in their sandbox account from a written brief, reusing the phase labs, then present the architecture and its monthly cost as they would to a manager.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Brief & Build	Read the brief, draw the architecture, then deploy the VPC, EC2 Auto Scaling group behind an Application Load Balancer, RDS database and S3 bucket	Project	3 h
4.2	Monitor & Estimate	CloudWatch alarms, a budget, a cost estimate with the AWS Pricing Calculator, architecture diagram	Project	1.5 h
4.3	Demo & Review	Live demo with an instance failure, cost presentation, teardown; questions and feedback	Project	1.5 h

FINAL PROJECT

A Highly Available Web Application on AWS

A web application that survives the loss of a server: a VPC across two Availability Zones, an Auto Scaling group behind an Application Load Balancer, an RDS database, static content on S3, CloudWatch alarms and a budget, presented with its monthly cost estimate.

STACK AWS: VPC, EC2, Auto Scaling, Application Load Balancer, RDS, S3, CloudWatch, AWS Budgets, AWS Pricing Calculator

DELIVERABLE Working deployment, architecture diagram, cost estimate, live demo